

PROVENANCE AND EVIDENCE SUPPLEMENT FOR POSITIVE-PARAMETER TWO-END SPATIAL DYNAMICS IN THE VAN DER POL REACTION-DIFFUSION SYSTEM

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1. PURPOSE AND RELEASE IDENTITY

This supplement records evidence status, frozen-source identities, theorem dependencies, and replay information for the companion paper. It is not a second mathematical proof and does not enlarge the theorem. The stable public release is

[https://github.com/h-lu/rfsn-ii-positive-parameter-pde/
releases/tag/vdp-companion-v1](https://github.com/h-lu/rfsn-ii-positive-parameter-pde/releases/tag/vdp-companion-v1).

The release tag identifies the application manuscript, the detailed V1–V7 proof archive, and the frozen input snapshot as one versioned object. The read-only upstream repository was not modified.

The companion theorem is conditional on Hypothesis H. Its analytic application conclusions are existential on a nonempty compact positive annulus. The proposed numerical Issue #7 box is not used as a theorem box.

2. FROZEN SOURCE

The frozen source is H. Lu, *First returns, singular exits, and action finite parts near a reversible Hamiltonian saddle-focus*, originally stored in the historical repository `h-lu/reversible-rfsn-ii-waves`. That repository is not the public locator for this supplement; the inspectable copy is the exact snapshot inside the application release. Its upstream commit is

`d54add098545063d5efe8f1d6f062d4cfc116a0d`

and tree

`b9cd34b1bae1c29bdd722d9ed3c33402c4b3ee89`.

The public application release contains an exact `git archive` extraction of the complete frozen `papers/paper-a/manuscript/` and `validation/` directories at

`frozen-imports/rfsn-ii-d54add098545063d5efe8f1d6f062d4cfc116a0d/
source/`.

No file below that directory was edited after extraction.

The principal SHA-256 identities are:

Date: August 28, 2026.

frozen file	SHA-256
manuscript source	0baf6335aad72d5893479d8876d2613671ecb8ac2ccd73664405dea4381e6a20
manuscript PDF	67888cf8b61b34c923cf55bd69ee41cab69493be6fc275533c8f5a074f1e96c5
replay manifest	15905f5a20b24a0ae0d298d9d14aa940177a006cc2cce3f2a39fd2e1cf4dac9b
environment lock	6240ebdbf0f296738534c07a33aea40202883f6abf37e1ff28e43dad47aa0cba
core homoclinic certificate	ed0f9f58f8ba5f1d5c36dc7c3a72bb725599c4172a3cd610d890b88699fecfbd
algebraic-anchor certificate	60882ee1d3b2b18264b85764288505ae8b47d00bc826a2bddec152898f690fbe
future-target certificate	88fa64035bb4352f5e25aa8d1627b191936264958c125dface59c5a767f6b3ce
pole-gate certificate	7f325a87810f8b0dda2542aed90263b39f9a9f4bd2e8ebb3abcc238d032eb6e2

The machine-readable copy of this table is `RELEASE_MANIFEST.json` beside the frozen snapshot README.

3. IMPORTED CLAUSES AND EXCLUSIONS

Only the following frozen clauses enter Hypothesis H.

- (1) The core system, one selected symmetric homoclinic, transversality modulo flow in the regular zero-energy hypersurface, its true unstable source graph, and the complete finite central physical event block.
- (2) The singular algebraic comparison sheet, source-incidence row, and endpoint-normalized Jost/adjoint row with exchange pairing $144\sqrt{3}$.
- (3) The analytic fixed-system saddle passage, endpoint and matching equations, winding selector, completed cross forms, clean first-exit family, exact section/gauge covariance, finite-atlas covariance, and terminal coding theorem.

The following frozen conclusions are not imported:

- either concrete singular end as a positive-parameter van der Pol end;
- a positive-parameter pole or algebraic compactification;
- a positive-parameter $K_2 \rightarrow K_1 \rightarrow$ outer matching theorem;
- a positive-parameter terminal counterterm or finite part;
- a global exact saddle chart or one parameter-global winding alphabet;
- temporal PDE stability, temporal chaos, Turing selection, canard identification, or experiment.

Thus the two positive ends, matching bridge, finite parts, event refinements, and translation to the original PDE are application results, not frozen imports.

4. CLAIM REGISTER CROSSWALK

The status column below reproduces the authoritative register verbatim. The dependency column records where a proved application statement uses Hypothesis H or an earlier H-dependent statement. Thus *Proved* is a proof status, not a claim that the frozen certificates were independently replayed by this repository.

ID	statement	status	dependency
V1	physical reversible exact Hamiltonian structure and primitive	Derived	direct calculation
V2	central saddle-focus, selected homoclinic, exact passage, and finite events	Proved	H_{core} and passage module
V3	genuine positive pole, open basin, labels, and pole action finite part	Proved	V2 pole window
V4	positive outer future-staying hypersurface and algebraic asymptotics	Proved	direct full-PDE construction
V5	$K_2 \rightarrow K_1 \rightarrow$ outer matching and moving-cut covariance	Proved	V2, V4, H_{Jost}
V5A	reference-subtracted algebraic length/action finite parts	Proved	matched V5 arrival
V6	exhaustive physical high-winding return/first-exit relation	Proved	V2–V5A and H_{ret}
V7	periodic, multipulse, and aperiodic stationary PDE patterns	Proved	V6 and coding module
T2G	one global exact saddle chart and global winding alphabet	Proposed	not used
S1	temporal spectral or nonlinear stability	Deferred	separate problem
E1	calibrated experimental observation	Deferred	separate problem

5. APPLICATION PROOF LOCATIONS

The reader-facing paper compresses the proof into four mathematical sections. The complete coordinate-level arguments remain in the versioned archive:

output	repository proof file
physical model, clocks, Hamiltonian	van-der-pol/HAMILTONIAN_CHECK.md;
frozen core statement	van-der-pol/MODEL_AND_CENTRAL_CHART.md
central continuation	van-der-pol/CENTRAL_CORE_IMPORT.md
positive pole terminal	van-der-pol/CENTRAL_CONTINUATION.md
outer future-staying graph	van-der-pol/POSITIVE_POLE_FINITE_PART.md
relative corner invariant manifold	van-der-pol/OUTER_FUTURE_STAYING.md
central-outer matching	theory/RELATIVE_OVERFLOWING_NHIM.md
algebraic finite parts	van-der-pol/CENTRAL_OUTER_MATCHING.md
frozen return/coding statement	van-der-pol/OUTER_ALGEBRAIC_FINITE_PART.md
conditional compact-family transfer	van-der-pol/RETURN_EXIT_CODING_IMPORT.md
physical finite-atlas descent	theory/COMPACT_FAMILY_FIRST_HIT_THEOREM.md
first events, actions, and PDE coding	theory/FINITE_MARKED_ATLAS_DESCENT.md
arrow-by-arrow publication audit	van-der-pol/TWO_END_RETURN_EXIT_AND_PDE.md
	proof-audit/VDP_PUBLICATION_PROOF_AUDIT_2026-08-28.md

6. REPLAY INSTRUCTIONS AND EVIDENCE LEVEL

In a full Git clone of the application repository checked out at the release tag, change to the frozen `source/` directory and run

```
python3 validation/replay_all.py \
  --profile main-theorem \
  --dry-run \
  --report /tmp/rfsn-main-theorem-dry-run.json
```

checks the manifest and frozen input hashes. Its successful status is

PASS-DRY-RUN-NO-CERTIFICATES-EXECUTED.

By design, this is not a mathematical replay. The frozen script is fail-closed and invokes Git, so this command is not advertised for a GitHub source ZIP or for a standalone copy without `.git`. Its report field `git_commit` identifies the enclosing application checkout; the upstream frozen identity is the `upstream_commit` in `RELEASE_MANIFEST.json`.

A full replay uses the same command without `-dry-run`, supplies the CAPD source and `capd-config` paths, and follows the compiler and FILIB constraints in the frozen environment lock. The interval computations use outward-rounded CAPD/FILIB enclosures; see the frozen validation README and the software records in the source package. The package is included so that the source claim can be inspected and replayed. This release does not claim that a second machine replay was performed by the application project.

The application repository also contains an unfinished explicit positive-box validation programme under `validation/rigorous/`. Its aggregate status remains `INCONCLUSIVE`, `claim_bearing=false`. Its development environment lock is distinct from the frozen core environment above and is not evidence for the existential annulus in the paper.

7. REVIEW AND LICENSING STATUS

The companion was subjected to mathematical seam audits, rendered-PDF checks, and a separate AI cold-reading task. These checks are not a substitute for an independent human expert review. A neutral expert packet is included beside the manuscript; no human reading is claimed until a named reader has actually returned a report.

Neither the application repository nor the frozen source contained a license file at the time of this release. The snapshot therefore preserves public inspection and citation access without inventing reuse rights. A future license decision belongs to the repository owner and is not a mathematical claim of this supplement.

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